

MATH 55 SECOND MIDTERM EXAM, PROF. SRIVASTAVA
APRIL 9, 2026, 3:40PM-5:00PM, VLSB 2050.

Name: _____

SID: _____

GSI: _____

NAME OF THE STUDENT TO YOUR LEFT: _____

NAME OF THE STUDENT TO YOUR RIGHT: _____

INSTRUCTIONS: Write all answers clearly in the provided space. This exam includes some space for scratch work which will not be graded. **Do not unstaple the exam.** Write your name and SID on every page. Show your work — numerical answers without justification will be considered suspicious and will not be given full credit. Write proofs in complete English sentences. Calculators, phones, cheat sheets, textbooks, and your own scratch paper are not allowed.

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Sign here: _____

Question	Points
1	8
2	6
3	6
4	8
5	8
6	6
7	8
Total:	50

Do not turn over this page until your instructor tells you to do so.
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Name and SID: _____

1. Select true or false for each of the following. There is no need to provide an explanation.

(a) (2 points) For all $n \geq 1$, $\binom{2n}{n} \leq \binom{3n}{n}$.

☐ True

☐ False

Solution: True. Since

$$\frac{\binom{3n}{n}}{\binom{2n}{n}} = \frac{(3n)!n!}{(2n)!(2n)!} = \prod_{i=1}^n \frac{2n+i}{n+i} > 1,$$

we have $\binom{2n}{n} \leq \binom{3n}{n}$.

Alternatively, here is a combinatorial proof. Let

$$A = \{S \subseteq [2n] : |S| = n\} \quad \text{and} \quad B = \{T \subseteq [3n] : |T| = n\}.$$

Then $|A| = \binom{2n}{n}$ and $|B| = \binom{3n}{n}$. Since $[2n] \subseteq [3n]$, every set in A is also a set in B , so $A \subseteq B$. Therefore $|A| \leq |B|$, i.e.

$$\binom{2n}{n} \leq \binom{3n}{n}.$$

(b) (2 points) If G is a simple graph with n vertices and $n - 1$ edges, then G is a tree.

☐ True

☐ False

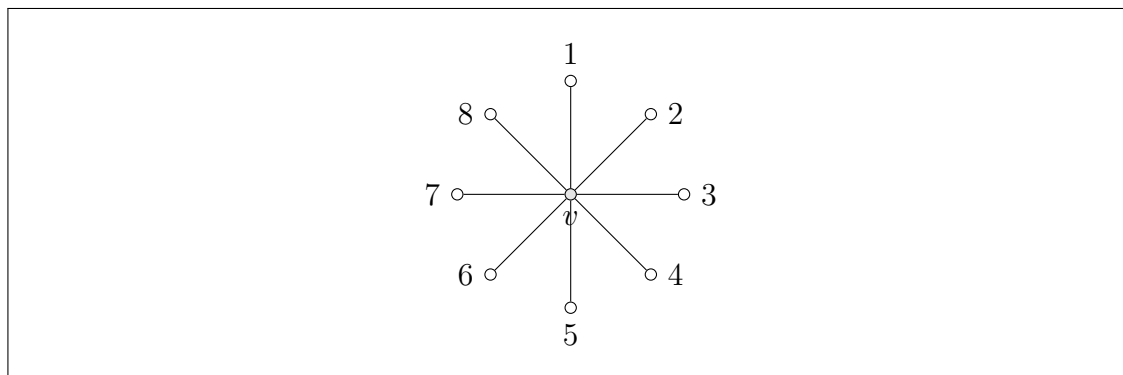
Solution: False. A triangle together with an isolated vertex has $n = 4$ vertices and $3 = n - 1$ edges, but it is not a tree.

(c) (2 points) If G is a graph with a vertex of degree 8, then G cannot be 2-colorable.

☐ True

☐ False

Solution: False. The star graph $K_{1,8}$ has a vertex of degree 8 and is 2-colorable. For example, color the center vertex v with color 1, and color each of the leaves $1, 2, \dots, 8$ with color 2.



- (d) (2 points) If G and H are isomorphic graphs, then they must have the same number of connected components.

☐ True

☐ False

Solution: True. Isomorphisms preserve adjacency, hence they preserve paths, connectedness, and the number of connected components.

2. (6 points) Prove by induction that $6 \mid n^3 - n$ for all $n \geq 1$.

Solution: Let $P(n)$ be the statement that $6 \mid n^3 - n$.

Base case: when $n = 1$, we have $1^3 - 1 = 0$, so $6 \mid 0$.

Inductive step: let $k \geq 1$ be arbitrary, and assume $P(k)$ holds. Then

$$\begin{aligned}(k+1)^3 - (k+1) &= k^3 + 3k^2 + 3k + 1 - k - 1 \\ &= (k^3 - k) + 3k(k+1).\end{aligned}$$

By the inductive hypothesis, $6 \mid (k^3 - k)$. Also, one of k and $k+1$ is even, so $2 \mid k(k+1)$. Therefore $2 \mid 3k(k+1)$. On the other hand, $3 \mid 3k(k+1)$ trivially. Since 2 and 3 are distinct primes, the Fundamental Theorem of Arithmetic implies that $2 \cdot 3 = 6$ divides $3k(k+1)$. Hence $6 \mid ((k+1)^3 - (k+1))$.

Thus $P(k+1)$ holds, and by induction $6 \mid n^3 - n$ for all $n \geq 1$.

3. A tech company requires employees to generate a secure 8-character password. The password must be made using only uppercase English letters (26 options) and digits (10 options).

- (a) (2 points) How many 8-character passwords can be created if there are no restrictions?

Solution: Each of the 8 positions has $26 + 10 = 36$ choices. By the product rule, the total number is

$$36^8.$$

- (b) (4 points) How many 8-character passwords contain at least three digits?

Solution: We can count the complement. There are 36^8 total passwords. The ones with fewer than 3 digits are the passwords with exactly 0, 1, or 2 digits:

$$\binom{8}{0}10^026^8 + \binom{8}{1}10^126^7 + \binom{8}{2}10^226^6.$$

Each term comes from the product rule, and the three cases are added by the sum rule. So the number with at least 3 digits is

$$36^8 - \left(\binom{8}{0}26^8 + \binom{8}{1}10 \cdot 26^7 + \binom{8}{2}10^226^6 \right).$$

Alternatively, for each $j = 3, 4, \dots, 8$, choose the j positions occupied by digits, then fill those positions with digits and the remaining positions with letters. For a fixed j , this gives

$$\binom{8}{j}10^j26^{8-j}$$

passwords by the product rule. Summing over all possible values $j = 3, 4, \dots, 8$, the total number is

$$\sum_{j=3}^8 \binom{8}{j}10^j26^{8-j},$$

by the sum rule.

4. Let $f(n)$ denote the number of ways to tile a $1 \times n$ strip using 1×1 , 1×2 , and 1×3 tiles, as illustrated below.



1×1



1×2



1×3

- (a) (4 points) Compute $f(1)$, $f(2)$, and $f(3)$, and find a recurrence relation for $f(n)$ when $n \geq 4$. Justify your recurrence.

Solution: We have

$$f(1) = 1, \quad f(2) = 2, \quad f(3) = 4.$$

For $n \geq 4$, consider the first tile in a tiling. If it has length 1, 2, or 3, then the rest of the strip can be tiled in $f(n-1)$, $f(n-2)$, or $f(n-3)$ ways respectively. These cases are disjoint and exhaustive, so

$$f(n) = f(n-1) + f(n-2) + f(n-3), \quad n \geq 4.$$

- (b) (4 points) Prove by strong induction that $f(n) < 2^n$ for all $n \geq 1$.

Solution: Let $P(n)$ be the statement that $f(n) < 2^n$.

Base cases:

$$f(1) = 1 < 2, \quad f(2) = 2 < 4, \quad f(3) = 4 < 8.$$

Inductive step: assume $k \geq 3$ and assume $P(j)$ holds for all $1 \leq j \leq k$. Then

$$f(k+1) = f(k) + f(k-1) + f(k-2)$$

by part (a). By the strong inductive hypothesis,

$$f(k) < 2^k, \quad f(k-1) < 2^{k-1}, \quad f(k-2) < 2^{k-2}.$$

So

$$f(k+1) < 2^k + 2^{k-1} + 2^{k-2} = 7 \cdot 2^{k-2} < 8 \cdot 2^{k-2} = 2^{k+1}.$$

Hence $P(k+1)$ holds.

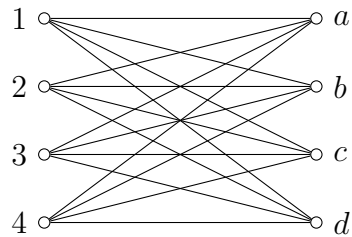
By strong induction, $f(n) < 2^n$ for all $n \geq 1$.

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5. Consider the graph $G = (L \cup R, E)$ where $L = \{1, 2, 3, 4\}$ and $R = \{a, b, c, d\}$ and $E = \{\{x, y\} : x \in L, y \in R\}$.

(a) (1 point) Draw a picture of G .

Solution: This is the complete bipartite graph $K_{4,4}$.



(b) (2 points) Is G Eulerian? Justify your answer.

Solution: Yes. The graph is connected, and every vertex has degree 4, which is even. Therefore G has an Eulerian circuit.

(c) (2 points) Is G 2-colorable? Justify your answer.

Solution: Yes. Color all vertices in L one color and all vertices in R the other color. Since every edge goes from L to R , this is a valid 2-coloring.

- (d) (3 points) Find the number of simple paths of length 4 in G starting at vertex 1. Justify your answer.

Solution: Since every edge of G has one endpoint in L and one endpoint in R , there are no edges joining two vertices of L and no edges joining two vertices of R . Therefore, whenever a path uses an edge, it must move from a vertex in L to a vertex in R or from a vertex in R to a vertex in L . In particular, any path starting at $1 \in L$ must alternate between vertices in L and vertices in R .

So a simple path of length 4 starting at 1 visits vertices:

$$1, r_1, \ell_1, r_2, \ell_2$$

where $r_1, r_2 \in R$ are distinct and $\ell_1, \ell_2 \in L - \{1\}$ are distinct.

There are 4 choices for r_1 , since r_1 can be any vertex of $R = \{a, b, c, d\}$. There are then 3 choices for ℓ_1 , since ℓ_1 must be a vertex of $L - \{1\} = \{2, 3, 4\}$. Next there are 3 choices for r_2 , since r_2 can be any vertex of R except the already used vertex r_1 ; it must be different from r_1 because the path is simple. Finally there are 2 choices for ℓ_2 , since ℓ_2 must be chosen from $\{2, 3, 4\}$ but must be different from the already used vertex ℓ_1 .

By the product rule, the total number is

$$4 \cdot 3 \cdot 3 \cdot 2 = 72.$$

6. (6 points) Prove that if G is a simple graph which contains at least two distinct spanning trees, then G must contain a cycle.

Solution: Let T_1 and T_2 be two distinct spanning trees of G . Since they are distinct and $|T_1| = |T_2| = |V| - 1$ because every spanning tree has $|V| - 1$ edges, we can choose an edge $e \in T_1$ such that $e \notin T_2$. Because T_2 is a tree there is a unique simple path in T_2 from u to v . Adding the edge e to that path creates a cycle. All of those edges lie in G , so G contains a cycle.

7. (a) (5 points) Prove that if G is a graph with $n \geq 2$ vertices and exactly two vertices x and y of odd degree, then there must be a path in G connecting x and y .

Solution: Let C be the connected component containing x . Degrees computed in C are the same as degrees computed in G , since there are no edges leaving a connected component.

By the Handshaking Lemma, the number of odd-degree vertices in the graph C must be even. Since x is odd, there must be at least one more odd vertex in C . But the only odd vertices in all of G are x and y , so that second odd vertex must be y .

Therefore x and y lie in the same connected component, so there is a path connecting them.

- (b) (3 points) Find an example of a graph G on 4 vertices such that (i) G has exactly two vertices of even degree, x and y , and (ii) there is no path in G between x and y .

Solution: Take

$$V = \{x, y, a, b\} \quad \text{and} \quad E = \{\{a, b\}\}.$$

Then $\deg(x) = 0$ and $\deg(y) = 0$, so x and y are even-degree vertices. Also $\deg(a) = \deg(b) = 1$, so these are the only two even-degree vertices. Since x and y are isolated, there is no path between them.

Graph Theory Definitions

A **graph** is a pair $G = (V, E)$ where V is a finite set of **vertices** and E is a finite multiset of 2-element subsets of V , called **edges**. If E has repeated elements G is called a **multigraph**, otherwise it is called a **simple graph**. Two vertices $x, y \in V$ are **adjacent** if $\{x, y\} \in E$. An edge $e = \{x, y\}$ is said to be **incident with** x and y . The **degree** of a vertex x is the number of edges incident with it.

A **path** of length k in G is an alternating sequence of vertices and edges

$$x_0, e_1, x_1, e_2, \dots, x_{k-1}, e_k, x_k$$

where $e_i = \{x_{i-1}, x_i\} \in E$ for all $i = 1 \dots k$. The path is said to **connect** x_0, x_k , **visit** the vertices $x_0, \dots, x_k$, and **traverse** the edges $e_1, \dots, e_k$. A path is called **simple** if it contains no repeated vertices. If $x_0 = x_k$ then the path is called a **circuit**. A graph is **connected** if it contains a path between every pair of distinct vertices.

A **k -coloring** of G is a function $f : V \rightarrow \{1, \dots, k\}$ such that $f(x) \neq f(y)$ whenever x is adjacent to y . A graph is **k -colorable** if it has a k -coloring.

A graph $H = (W, F)$ is a **subgraph** of G if $W \subseteq V$ and $F \subseteq E$. If $W = V$ then H is called a **spanning** subgraph of G . A **connected component** of G is a maximal connected subgraph of G , i.e. a subgraph $H \subseteq G$ such that H is connected and adding any vertices or edges to H would make it disconnected.

A **cycle** is a circuit with no repeated edges and no repeated vertices, except the endpoints. A graph is called **acyclic** if it does not contain a cycle as a subgraph. A connected acyclic graph is called a **tree**.

An **Eulerian circuit** of a connected multigraph G is a circuit which traverses every edge of G exactly once.